

Controlling Cross-Content Motion Style Transfer via Statistical Style Difference

Usfita Kiftiyani
Kyung Hee University
Republic of Korea
kiftiyani@khu.ac.kr

Seungkyu Lee
Kyung Hee University
Republic of Korea
seungkyu@khu.ac.kr

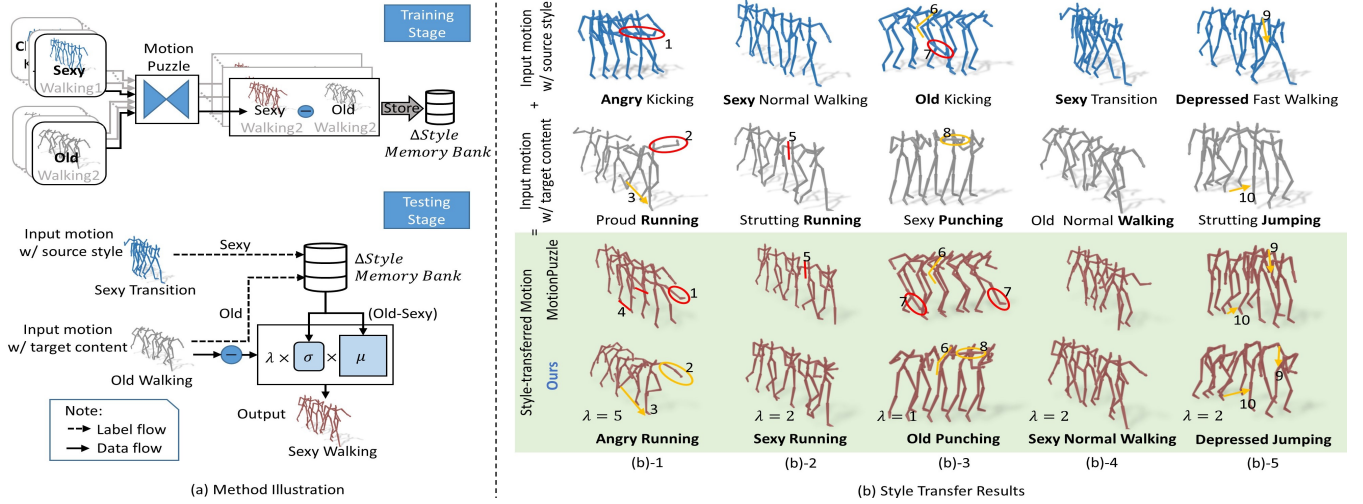


Figure 1: (a) Overview of the proposed method. (b) Qualitative comparison between our method and MotionPuzzle [Jang et al. 2022] on Xia [Xia et al. 2015] dataset. Red color indicate wrong transfers and yellow color for correct transfers.

Abstract

Decomposing human body motion content (such as walking, jumping, punching) and style (such as angry, old, proud) is a challenging task in style transfer, often causing unrealistic content-style combinations. This study demonstrates a straightforward method obtaining statistical style differences ($\Delta Style$) which in turn effectively replace original style from a motion content to another target style. The strength of transferred style is adjusted using a predefined parameter of our model. Experimental results show that proposed $\Delta Style$ significantly improves style transfer quality particularly in cross-content scenarios.

CCS Concepts

• Computing methodologies → Motion processing.

Keywords

Motion Style Transfer, Style Controller, Motion Decomposition

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1 Introduction

Typical approaches on style transfer involve extracting style component from a source motion and transferring it to a target content. MotionPuzzle [Jang et al. 2022] proposes BP-StyleNet to stylize motion at body parts. However, the method does not explicitly decompose body motion into "content" and "style" components in its method. This leads to a problem that both content and style of source motion are transferred to target motion showing unexpected content movement in the result.

We aim to explicitly decompose a motion into content and style components considering their labels (e.g., "childlike" as the style and "kicking" as the content). To this end, there are two issues to address. First, only style of source motion has to be transferred removing content of the source motion not to affect to target motion. For example, red circles of Figure 1(b)-3 shows content of source motion is unexpectedly transferred to target motion in MotionPuzzle [Jang et al. 2022] results. Second issue is that a style does not mean any predetermined motion but it can be expressed with diverse movements of different body parts. For instance, a "proud" style expression involves sometimes exaggerated hand gesture in "proud

running" or just subtle hand movements in "proud kicking". This variety in style expression make clean style transfer difficult.

To address the issues, we propose a straightforward method that utilizes statistical differences in style, called $\Delta Style$, to control the transfer of style information, especially in cross-content scenarios. By controlling the transferred style information, our approach effectively suppresses unwanted content movement transferring and adjusts the strength of style transfer amount thanks to the statistical information of each style obtained in $\Delta Style$.

2 Method

Our proposed method first build $\Delta StyleMemoryBank$ that accumulates pairwise statistical style difference information. And then style transfer of input motion (testing stage in Figure 1(a)) is performed by referring $\Delta Style$ between current style of input motion ("old" in "old walking") and new style ("sexy" in "sexy transition") to transfer.

As illustrated in Figure 1(a), to obtain style only information during subtraction, different style motions of the same content label and instance are needed in training stage. We achieve this by embedding style input motion from an instance (illustrated with "sexy walking1") to another instance with the same content label but different style motion (illustrated with "old walking2", i.e., "walking1" and "walking2" indicate the same content label from different instances), resulting in "sexy walking2". The style difference is then calculated between the pairs of styles within the same instance ("sexy walking2" and "old walking2") and stored in the $\Delta StyleMemoryBank$. Note that each pairs of style in the memory bank consist of multiple instances that will be referred in the testing stage by taking the statistical information of all instances of the corresponding pairs. As a result, the $\Delta StyleMemoryBank$ grows quadratically w.r.t. the number of styles, limiting its implementation to the availability of style-labeled data.

In our style Transfer Mechanism in the testing stage, as shown in the bottom part of Figure 1(a), we first refer the $\Delta StyleMemoryBank$ to extract the statistical components—namely mean $\mu_{C_s, S_s} \in \mathbb{R}^{C \times V \times T}$ and variance $\sigma_{C_s, S_s} \in \mathbb{R}^1$ —of corresponding pairs of style from the input motions, illustrated by (old-sexy) of $(C_s - S_s)$. Then the style transfer conducted as follows,

$$M_{S,C} = M_C - \lambda \times \sigma_{C_s, S_s} \times \mu_{C_s, S_s} \quad (1)$$

Given input motion with target content $M_C \in \mathbb{R}^{C \times V \times T}$, illustrated in the testing stage of Figure 1(a) with "old walking". The second term of Eq. 1 aims to remove the style information from the target content C_s (old) and adding the style information corresponding to the source style S_s (sexy). Finally, we control the amount of transferred style information using a predefined parameter $\lambda \in \mathbb{R}^1$. Then resulting the style-transferred motion $M_{S,C}$ (sexy walking).

3 Experimental Results and Discussion

To demonstrate the performance of our proposed method, we conduct experimental evaluations on Xia [Xia et al. 2015] Dataset and compare our results with those from MotionPuzzle [Jang et al. 2022].

Our method improve MotionPuzzle limitation on suppressing unexpected content component of the input motion with source style shown in 1st and 7th red circles of Figure 1(b)-1 and (b)-3,

Table 1: Quantitative Evaluation on Xia Dataset.

Methods	FMD↓	CRA↑ (%)	SRA↑ (%)
Real motions		75.52	51.40
MotionPuzzle	30.52 ± 0.5	20.29 ± 0.38	57.12 ± 0.27
Ours ($\lambda = 1$)	0.886 ± 0.124	59.38 ± 0.46	32.39 ± 0.31
Ours ($\lambda = 2$)	1.62 ± 0.097	51.93 ± 0.45	28.59 ± 0.72
Ours ($\lambda = 3$)	3.25 ± 0.149	47.92 ± 0.27	24.94 ± 0.58

The FMD, CRA, and SRA scores are evaluated on 2,500 samples generated by each method per trial. The table reports the mean ($\pm$ standard deviation) for each metric over 5 trials. Note that we use our own metric calculations based on [Jang et al. 2022], so results may differ from the original MotionPuzzle. All input samples are interpolated to 100 frames for a fair comparison.

respectively. We demonstrate the effect of our λ adjustment in Figure 1(b)-1. By using a larger λ value, we show that the exaggerated hand movement in red circle numbered 2 can be reduced resulting in yellow circle numbered 2. In this instance, MotionPuzzle also effectively reduces these issues due to its learning capability.

Finally, we conduct quantitative evaluation following the evaluation scheme in [Jang et al. 2022] by using three metrics; Fréchet Motion Distance (FMD), content recognition accuracy (CRA), and style recognition accuracy (SRA). Table 1 shows that our method improves the generation results to be more realistic, with FMD score of 0.886 and also reflected in the CRA score, where our method effectively preserves the content components of the target content. The evaluation model struggles to recognize subtle style differences within the same content due to its naive implementation. This results in a lower SRA score for real motion data compared to MotionPuzzle. Similarly, our method also yields a lower SRA, primarily due to the influence of the parameter λ . The λ value varies depending on the motion intensity in the input, as illustrated in Figure 1(b)-1, a high λ value is required to remove the prominent style in the input content, while a lower λ value is sufficient for other cases in Figure 1(b). This highlights the limitation of using a predefined parameter λ in our method.

4 Conclusion

This study presents a straightforward approach to style transfer that leverages statistical differences between styles to remove unexpected content component transfer from the source. We introduce a predefined parameter, λ , to control the transferred style strength. Our method outperforms MotionPuzzle on Xia dataset.

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